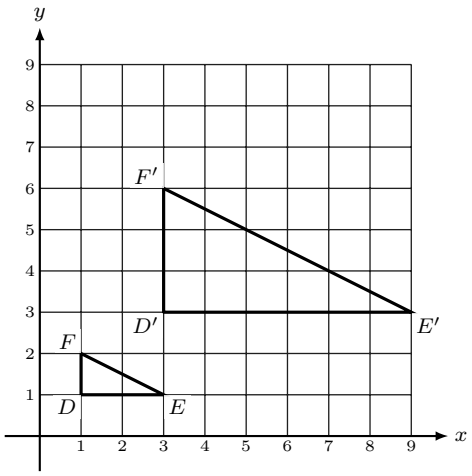
Artifact 6 – Algebraic Rules of Dilations

Driving Question: How can any dilation be written as an algebraic rule?

Example 1



Pre-Image	Image

$(x, y) \rightarrow \left(\quad , \quad\right)$

Example 2

Triangle JKL is dilated with the origin as the center of dilation to create triangle $J'K'L'$.

$\triangle JKL$	$\triangle J'K'L'$
$(-6,\ 8)$	$(-3,\ 4)$
$(4,\ -2)$	$(2,\ -1)$
$(10,\ 6)$	$(5,\ 3)$

$(x, y) \rightarrow \left(\quad , \quad\right)$

Example 3

Point $P(8,\ -6)$ is dilated with the origin as the center of dilation. Its image P' is located at $(12,\ -9)$.

$(x, y) \rightarrow \left(\quad , \quad\right)$

Vocabulary

Algebraic representation (rule) of a dilation
Variable scale factor (k)
Similarity

Notes to Forgetful Self

Scale factor	Rule	Scale factor	Rule
2	$(x, y) \rightarrow \left(\quad , \quad\right)$	$\frac{2}{3}$	$(x, y) \rightarrow \left(\quad , \quad\right)$
3	$(x, y) \rightarrow \left(\quad , \quad\right)$	$\frac{5}{2}$	$(x, y) \rightarrow \left(\quad , \quad\right)$
$\frac{1}{2}$	$(x, y) \rightarrow \left(\quad , \quad\right)$	k (any)	$(x, y) \rightarrow \left(\quad , \quad\right)$

Not a dilation.

$(x, y) \rightarrow (x + 2,\ y + 2)$ is a

$(x, y) \rightarrow (-x,\ y)$ is a